Did Working with Women Change the Workplace? Evidence from Wartime Tokyo *

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Abstract

How do organizations replace workers lost to exogenous exits? I study this question using military conscription during World War II, which removed workers from the Tokyo civil service in a manner that was largely random conditional on occupational role. Using a novel personnel database covering approximately 40,000 civil servants per year from 1928 to 1960, I find that impacted offices received three times as many internal transfers as external hires. Offices mitigated disruption by promoting lower-ranked workers and offering bonus salaries to retain remaining staff. The central human resource team facilitated reallocation by reassigning workers from offices with redundant staff, and these donor offices did not increase their own hiring — contradicting theories of vacancy chains. External hires were predominantly male and concentrated in lower-ranked positions, with no effect on the share of female workers. These results reveal that organizations absorb labor supply shocks primarily through their internal labor markets, and that internal reallocation — not external hiring — determines which jobs become available to new entrants.

Keywords: Female representation, Long-term effects, Personnel economics

JEL Codes: J71, M51, M54, N45

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1 Introduction

How do workplaces respond to exits of their staff caused by military drafting? Wars mobilize a vast number of workers across sectors through conscription of the active labor force into the military, as well as voluntary exits caused by the expansionary effect of government spending on defense sectors. A vast literature has discussed how displacement accelerated the integration of underrepresented populations into the workforce[1][2], but the exact adaptation of the labor demand side to military drafting remains untouched. This is staggering, as an employer's hiring pool should be a joint pool between the internal and external labor market. In particular, military spending often overheats labor markets, and the hiring costs for new workers should induce employers to look for replacements within their organizations or manage the exit by reallocating their existing contracted workers. This shortcoming limits our understanding of (1) the types of jobs that become available to external workers during war periods, and (2) how those jobs are generated through workplaces' efforts to mitigate disruption by managing their existing human resources.

I document the reactions of workplaces to fill drafting-induced vacancies through recruiting from the internal and external labor market by assembling a novel database on civil servants in wartime Tokyo, which provides the positional information of the universe of its employees. The data allows me to test how managers filled vacancies under their supervision by promoting existing workers, poaching workers from adjacent offices, and hiring from external labor markets. I find that managers' efforts to fill empty positions varied by the type of job of the drafted workers, and that the worker pool from which managers drew replacements also varied by job type.

The publicly undisclosed drafting process removed workers from the Tokyo civil service, who were distributed across a large hierarchical, multi-layered system. The process and results of the conscription were hidden from the public to prevent eligible workers from avoiding enlistment in the military. Therefore, the distribution of draft-eligible employees across office units within the Tokyo civil service, especially after conditioning on occupational roles, was largely random. Addi-

[1]The heating of local labor markets further hindered managers' efforts to find replacements from external labor markets, as government spending to support the war overheated the labor market (**?**).

[2]Ferrara (2022) on Black labor supply, Acemoglu et al. (2004); **?**); **?** on female labor supply

tionally, the national civil servants were exempt from the drafting, and therefore the vacancy-filling responsibility was largely assigned to the manager of the office unit. As the Tokyo civil service employed a multi-layered hierarchical structure for providing public services to Tokyo's constituents, I identify the impact of drafting by comparing neighboring office units with differential numbers of drafted workers. The service consisted of nearly 12 departments (*kyoku*), each nesting multiple offices (*ka*), and each *ka* including multiple units (*kakari*). Within each *kakari*, employees were further separated into groups by role, ranging from engineering roles to administrative support roles. Office structures across *kakari* units within a *ka* were largely similar to each other, as they shared similar tasks to achieve the goals assigned to their affiliated *ka*. Using the military drafting-induced exits of employees, which the database identifies at the individual level, I estimate how the occupational groups within a *kakari* across its affiliated *ka* changed their composition of workers.

I find that drafting induces workplaces to find replacements from the internal labor market at 25% times lower the rate of external hires. A draft-induced exit increases the number of internal transfers into the office by 0.15 workers relative to the adjacent office and 0.21 workers from external markets. Alongside recruiting workers from internal offices, offices mitigated the disruption caused by drafting by promoting workers in lower-ranked positions and offering bonus salaries to the remaining staff to retain them. A one-unit increase in the number of drafted workers increased the probability of a worker in the same office being promoted by 8 percentage points. Additionally, a similar exit increased the number of existing staff on bonus salary schedules by 0.1 units. I find that retention probability for existing workers was higher in draft-impacted offices, but the estimate is statistically insignificant.

External hires induced by drafting were mainly men, with little effect on the number of women entering the civil service. A one-unit draft-induced exit caused an additional 0.2 male hires. The additional external recruiting induced by drafting was mainly concentrated in lower-ranked positions, with 0.01 additional women and 0.3 additional men. As a result, I do not find any improvement in the share of female workers. Instead, drafting reduced the average length of tenure of staff members in the impacted office unit.

I find suggestive evidence that military drafting accelerated the promotion of pre-existing staff. Workers from impacted offices were assigned to positions 0.1 standard deviations higher in rank on average than those in adjacent offices, though the estimate is not statistically significant. I do not find any impact on office composition in future assignments between workers from non-drafted and drafted offices.

The results underscore how workplace disruptions due to military drafting were met by workplaces seeking replacements from internal sources rather than the external market. The expertise of workers constitutes a valuable asset for workplaces, and replacing workers with external hires imposes an additional cost (**?**, **?**). The composition of the neighboring organizational structure predicts the success of finding a replacement to fill the role, as the central human resource management team can identify departments with unused employees and relocate them to the impacted office. Such adjustments do not generate any additional hiring at the selected office, as human resources chooses offices in a way that mitigates the overall disruption to the functioning of the workplace, contradicting theories of vacancy chains ().

The responses of workplaces to reallocate workers based on expertise did not create differential exposure to women for incumbent men who were not called up for drafting. The inflow of women into the civil service was mostly associated with voluntary exits of men who appear to have left the civil service to find other working opportunities outside the service. This was fueled by the temporary overheating of the labor market due to increased government spending to support the war. However, the drafting did increase hiring in low-ranked positions, and this may have had lasting effects on the managerial experiences of workers who were promoted as a result of drafting.

2 Background

2.1 Institutional Details of Tokyo Civil Service

The Tokyo public service, comprising the *Tokyo-Fu* and *Tokyo-Shi*, managed public good provision across geographically distinct areas of Tokyo. Although these bodies operated over separate regions, their main offices were centrally located in downtown Tokyo (Figure 1).

In 1944, under intensifying war pressures and facing repeated military defeats, the central government mandated the amalgamation of *Tokyo-Fu* and *Tokyo-Shi* into a single entity, named *Tokyo-To*, through a federal bill. This merger, driven by operational inefficiencies and resource allocation challenges during the war, resulted in significant layoffs, particularly affecting *Tokyo-Shi* employees.

Tokyo-To employed approximately 15,000 highly educated staff, with 30 percent hired through national exams and the remainder locally. The organizational structure was hierarchical, dividing departments based on the public services provided, with some departments split into as many as four layers. Employees were assigned an occupation, such as engineers, managers, clerks, or assistants, along with a classification reflecting their professional expertise and experience. These assignments were determined upon entry into the organization, although transitions across occupations occasionally occurred. The recruitment process for non-assistant employees was merit-based, relying on competitive examinations and internal assessments. Within the civil service, workers were further categorized into upper and lower classes, with managers representing the upper class and clerks or engineers forming the lower class. Leadership positions, such as department heads, were typically filled by personnel recruited from the central government. For many employees, the typical career path involved progressing step by step up the job ladder, with clerks serving as the standard entry point for civil servants aiming to advance. However, the level of competition for promotions among employees was likely influenced by the conscription of a significant portion of the workforce into military service, which temporarily reduced the pool of available candidates. This shift may have also led to substantial changes in both the promotion process and the speed of advancement for positions such as clerks and managerial staff. In fact, many staff members were conscripted as soldiers, resulting in a severe manpower shortage for managing wartime administration (Tokyo Metropolitan Government 2013).

2.2 Drafting Process of the Japanese Military Service

The Japanese military assembled its forces by calling active service troops and reservists, a process pivotal to our study's analysis of workforce dynamics during and post-conflict periods. The conscription system was designed to ensure a steady supply of personnel by mandating physical examinations for all eligible men upon reaching the age of 20, with those meeting the criteria randomly selected for service.

The drafting process, as outlined in Figure 3, was structured into several stages to efficiently classify conscripts based on their physical capabilities and the military's operational needs. Initially, all eligible men underwent a physical examination to determine their fitness for active duty. Those deemed fit were assigned to active service, where they served as soldiers for a fixed period. Upon completing active duty, these individuals transitioned to the reservist category, ensuring they remained available for military recall during national emergencies. For individuals who did not meet the physical standards for active duty, an alternative route, termed replacement service, was established. These men received training but were not immediately deployed, instead forming a reserve pool for supplementary support when required. Additionally, a subset of candidates was fully exempted from military obligations due to severe health conditions or other qualifying factors.

As the war intensified, particularly in the wake of Japan's challenges during the Second Sino-Japanese War (1937–1945), the drafting system underwent significant modifications to address the military's increasing personnel demands. In 1943, reserve obligations were extended: the period of reserve service following active duty increased from 5–6 years to 15 years, and the secondary reserve period was prolonged until the age of 45. These changes ensured that the military could rely on a broader pool of trained personnel over an extended timeframe, providing critical support for wartime operations.

2.3 The Impact of the War on Tokyo-To

The war significantly altered the workforce composition within *Tokyo-To*, particularly affecting female participation rates and the distribution of staff across various offices. The conscription system, which required eligible men to serve in the military, had profound implications for public institutions, especially in urban centers like Tokyo. Male workers in administrative, engineering, and clerical roles were not exempt from military obligations, leading to their systematic removal from public service positions as they were drafted into active or reserve duty. This mass conscription created substantial disruptions in the operations of public offices and left critical gaps in the workforce that urgently needed to be addressed.

During the war period, the entry of female workers also became a notable feature of *Tokyo-To's* workforce. Before the war, female employees were largely concentrated in specific departments and were almost absent from the core administrative divisions of the metropolitan government. Their roles were typically limited to auxiliary or supporting tasks, reflecting the gendered labor practices of the time. However, the rapid expansion of labor demand driven by conscription accelerated the entry of women into the workforce. Units were generally structured to include a manager, clerks or engineers, and assistants. Female workers were primarily hired as assistants to support unit leaders in their tasks, often in offices with a lower concentration of engineers. Their continuation and promotion within the organization depended heavily on the approval of their respective unit managers, with their relationships with male incumbents—such as direct subordinates to unit managers or co-workers to clerks and engineers—playing a significant role. In addition to conscription, the demands of wartime administration likely necessitated an increase in the number of staff employed by *Tokyo-To*. The rising responsibilities of the metropolitan government during the war made it essential to expand its workforce. Historical evidence suggest that both the city and the prefecture increased the number of employees and departments during this period (Tokyo Metropolitan Government 2013).

To compensate for the depletion of male workers, *Tokyo-To* rapidly increased the recruitment of women, who were not subject to military service. Women entered the public workforce in unprecedented numbers, particularly in assistant roles, as highlighted in Figure 5. Panel (a) illustrates the sharp increase in the population of female public civil servants during the World War II (WWII) period, marked by the shaded red area. The count of female employees rose significantly between

1938 and 1944, reflecting a direct response to the conscription of men. However, this trend reversed immediately after the war as male workers returned to their civilian roles, leading to a sharp decline in the number of female public civil servants.

Panel (b) of Figure 5 provides further insight by examining the share of women in the public sector workforce during the same period. The proportion of female public civil servants also increased markedly during WWII, albeit remaining relatively small compared to the overall workforce. The rise in this share highlights not only the increasing reliance on women but also the gendered constraints of wartime employment policies, which often relegated women to lower-ranking and temporary roles within the organizational hierarchy. This unequal distribution persisted even after the war, as men resumed their pre-war positions, and women were often pushed out of public service.

Table 1 shows the difference between offices in 1936 with and without employees that were drafted 2 years later. We find a statistically significant different between the share of female workers between offices categories. Workers in future drafted offices were more likely to be occupied with male. Since the Japanese military service only conscripted males, the difference in gender balance between offices validates the drafting pattern. The two office categories matches each other with respect to its share of new workers and share of higher-ranked workers. We fail to reject similarities between the at least at the 15th confidence level. Drafted and non-drafted employees worked in offices with nearly half of their peers hired at most 4 years before. Also they did not differ in the share of higher-class public servants that they reported to.

3 Data

Our primary data set comes from a historical archive of directory of staff of the Tokyo public servants. The archives include information about the name of the employee, their positions, and their home address. We infer the gender of the staff by using an algorithm trained on historical Japanese names. The personnel record covers from 1928 to 1958.

The directory of staff was published annually with a full list of their employees. The two co-governing institutions (*Tokyo-Shi* and *Tokyo-Fu*) both published separately until the merger of the two institutions in 1943. The Tokyo-Fu directory contains a list of the employees that was drafted into the war. Since we observe the positions of all the employees in a institution, we identify the drafted individual and their surrounding co-workers in a directory.

Our secondary data is the biographical directory of staff at 1942. The biography lists detailed information about higher ranked employees. It includes (1) Date of birth, (2) Prefecture of origin, (3) Education attainment (name of graduating institution) (4) Starting date of entering the public servant system, (5) hobbies, and (6) Physical address. However, the directory was not published annually like the directories, but only published in some years. Also only the Tokyo-Shi and Tokyo-To published biographical directories of their employees. In addition the biographical directory only reports the details for employees at least in positions higher than a clerk.

We develop an algorithm that integrates multiple neural networks and computer vision technologies to detect complex layout structures in archival records and extract categorized text data. Given that most modern optical character recognition (OCR) systems are primarily trained on the English language, we utilize a neural network-based OCR developed by the National Diet Library in Japan. This OCR is adept at identifying texts within a document page, which we further analyze using named entity recognition technology to categorize information based on content and layout. We apply this algorithm to a series of historical documents to augment the biographical details of employees. This process generates a comprehensive panel data set of public sector employees in historic Tokyo, detailing their internal assignments by occupation and rank, along with biographic details for a subset of these employees.

Our OCR works better for the recent years than the older years. Since the calligraphy of Japanese letters changed after the war, a modern OCR fails to accurately identify the letters in the directory. Eventhough we use the NDL OCR, which is trained on historical Japanese letters, due to the resolution of the scanned images, the OCR sometimes fail to identify the content. However, we di recover approximately 70% of employees in the historic era, and almost everyone for those in the more modern era (Figure 7).

We predict the gender of the employee by using an external named entity recognition algorithm that predicts the gender of the name. In contrast to Western names, Japanese names tend to reveal the gender of the employee easily. We measure the size and share of female workers of each office unit by aggregating over the pool of employees belonging to a given group. We measure the length of tenure for each employee that we identify in the directory by searching for the first and last year of the directory that we identify their name. Consequently, if an employee is missing in a directory between two directories in which the employee is detected, we treat the employee to be hired in the missing directory. We also infer the location of the office that an employee commuted to by referring to the name of the office unit.

The digitizization of the historical archival documents leaves us with a sample of 333,000 observations across 35 years. The data identifies 126,372 unique names and 15,406 female names.

4 Empirical Model

4.1 Drafting specification

We estimate the effect of military drafting on office-level outcomes using three specifications. The unit of observation is position $\times$ year (1938–1945), unless noted. All specifications include year fixed effects and either ka (section) or kyoku (department) fixed effects, plus position fixed effects. Standard errors are clustered at the office level.

Reallocation and hiring (Tables 3, 8, ??). Count outcomes (new hires, transfers in) are modeled with Poisson regression:

$$\mathbb{E}[Y_{j,p,t} \mid X] = \exp\left(\beta_1 D_{j,p,t} + \beta_2 D^{\text{adj}}_{j,p,t} + \beta_3 \log(M_{j,p,t} + 1) + \gamma + \delta_p + \lambda_t\right), \tag{1}$$

where $Y_{j,p,t}$ is the number of new hires or transfers into position p in office j in year t; $D_{j,p,t}$ is the number of male workers drafted from that position; $D^{\text{adj}}_{j,p,t}$ is the number drafted from adjacent positions within the same office; and $M_{j,p,t}$ is the cumulative male stock. The coefficient β_1 is a semi-elasticity: a one-unit increase in own drafts is associated with a $100 \times \beta_1\%$ change in expected hires or transfers.

Gender composition (Table ??). Female share among new hires or among all workers is modeled by OLS. The sample excludes *kanri* (higher-tier civil servant) positions. Treatment is either an extensive margin indicator (any drafted) or draft share (drafted male count / male baseline). We control for cumulative male baseline and the same fixed effects as above.

Postwar tenure (Table 10). For non-drafted workers observed in 1944, we regress postwar tenure (count of years appearing in civil service through 1955) on drafting exposure in their 1944 office. The unit of observation is the individual; treatment is any drafted or draft share; fixed effects are ka (or kyoku) and position as of 1944.

4.2 Identification: conditional independence

Causal inference requires that drafting be as good as random conditional on the fixed effects. Formally, we require that within ka $\times$ position (or kyoku $\times$ position) cells and by year, the number of workers drafted from a position is independent of potential outcomes and of other pre-treatment characteristics. This conditional independence assumption is plausible if drafting was determined primarily by national conscription rules (age, health, exemption status) rather than by office-specific factors that also predict hiring or tenure.

Table 2 provides evidence. Panel A shows an unconditional comparison of pre-war (1937–1942) office-year characteristics by whether the office was drafted in 1944. Unconditionally, drafted offices had a higher new-hire share and lower average rank, consistent with drafting being more likely where turnover was higher and workers were lower-tier (and thus less exempt). Panel B conditions on ka $\times$ modal occupation and year fixed effects, matching our main specifications. Within these cells, all six pre-war characteristics are uncorrelated with the drafted-in-1944 indicator (p-values 0.31–0.94). Conditional on organizational structure and occupational mix, drafting appears as good as random.

5 Results

5.1 Worker Reallocation in Response to Military Drafting

Table 3 reports Poisson estimates of how offices respond to losing male workers to military drafting. The unit of observation is position $\times$ kakari $\times$ year (1938–1945). Column 1 shows that each additional draft in the own position raises total transfers in by approximately 5.5% ($\beta = 0.055$, $p < 0.01$). Columns 2–3 decompose these transfers by origin: same-ka transfers (internal reallocation) increase by about 9.6% per draft, while different-ka transfers (external poaching) increase by 5.4%. Columns 4–5 further decompose external transfers by organizational distance: same-kyoku transfers rise by 5.3% and different-kyoku transfers by 4.7%, indicating that the response declines modestly with network distance. Table 8 reports new hiring. Column 1 shows that each draft raises new hiring by approximately 3.9%. Columns 2–3 report spillovers from adjacent drafting: drafts at different kakari within the same occupation raise new hiring by about 0.4% (column 2), and drafts at different occupations within the same kakari raise new hiring by about 0.3% (column 3). All specifications include year, ka, and position fixed effects; standard errors are clustered at the office level.

[*To fill in manually:* Interpretation, policy implication, or connection to prior work]

5.2 Gender Composition of the Hiring Response

Table **??** investigates whether drafting-induced vacancies are filled differently by gender. Columns 1–2 report Poisson estimates for female and male new hires separately. Male hiring increases significantly by 3.6% per draft ($\beta = 0.036$, $p < 0.01$), while the increase in female hiring is positive but imprecise (3.2%, $p > 0.10$). Columns 3–4 add an interaction with rank 1 (*yatoi*/temporary) status: the interaction on own drafts is large for female hires (12.8 percentage points) but statistically insignificant, suggesting that temporary positions may have absorbed more female hiring in response to drafts, though the evidence is inconclusive. Table **??** reports OLS estimates for female share of the workforce and female share of new hires by rank (R1 = temporary, R2 = regular).

The coefficients on the any-drafted indicator are small and statistically insignificant across all four specifications (female share of all workers at R1 and R2, and female share of new hires at R1 and R2). This indicates that while male hiring increases in both absolute and proportional terms, female hiring increases in absolute terms but does not systematically shift the gender composition of the workforce or of new hires.

[*To fill in manually:* Interpretation, policy implication, or connection to prior work]

5.3 Postwar Outcomes of 1944 Colleagues

Table 10 examines postwar outcomes (1946–1955) among non-drafted workers observed in 1944. Columns 1–3 report OLS estimates with z-scored rank change as the dependent variable (postwar average rank minus 1944 rank, standardized within era). The coefficient on the any-drafted indicator (column 1) is 0.058 (approximately 0.06 standard deviations) but imprecise. Column 2 uses draft share instead; the coefficient is 0.073 and similarly imprecise. Column 3 adds a gender interaction: the main effect of draft share remains positive (0.055), the female main effect is negative (-0.57), and the draft share $\times$ female interaction is large and positive (1.06), suggesting that women in more heavily drafted offices may have experienced larger rank gains than men, though the interaction is not statistically significant. Columns 4–5 examine postwar retention (count of years appearing in civil service through 1955): the coefficients on any drafted and draft share are positive (0.037 and 0.050) but imprecise. Columns 6–7 analyze the average female share of workers' destination office-positions in the postwar period; drafting exposure does not systematically predict the gender composition of postwar placements (coefficients near zero and insignificant). All specifications include ka and position fixed effects from 1944; standard errors are clustered at the office level.

[*To fill in manually:* Interpretation, policy implication, or connection to prior work]

5.4 Effect of Drafting on Office Size

Table **??** reports OLS estimates of how military drafting affects remaining office headcount. The unit of observation is kakari × position × year (1938–1945). The dependent variable is the number of non-drafted workers remaining in the office. Column 1 shows that each additional male drafted reduces remaining headcount. Columns 2–3 add controls. All specifications include ka, position, and year fixed effects; standard errors are clustered at the office level.

[*To fill in manually:* Coefficient interpretation and connection to replacement hiring results]

5.5 Characteristics of Donor Offices

Table **??** compares offices that transferred workers out against those that did not, conditional on ka × position fixed effects. The table reports differences in whether the office hired anyone new that year, average tenure length, and female share. These comparisons shed light on whether offices selected to surrender workers had distinct workforce characteristics, offering a test of vacancy-chain theories.

[*To fill in manually:* Interpret patterns and connect to theory]

5.6 Characteristics of Transferred Workers

Table **??** examines the individual characteristics of workers who were transferred out of donor offices. The regressions test whether transferred workers were disproportionately female, had different tenure lengths, or held different ranks (as a proxy for productivity), conditional on ka × position fixed effects.

Table **??** examines postwar outcomes (1946–1955) for workers who remained in donor offices compared to workers in adjacent kakari that did not experience transfers. The analysis tests for differences in tenure length and promotion trajectories.

[*To fill in manually:* Interpret transferred worker selection and postwar effects]

5.7 Women in the Civil Service, 1928–1955

Figure **??** plots the number of women employed in the Tokyo civil service from 1928 to 1955, distinguishing between *kanri* (, appointed officials) and *kouri* (, public employees). Female participation was effectively zero before 1937 and rose sharply during the war years as offices responded to labor shortages caused by military drafting. The postwar trajectory shows whether wartime gains in female employment persisted through the reform period.

[*To fill in manually:* Connect to main argument about persistence of wartime female inclusion]

5.8 Assignment of Women to Offices Post-Reform

Table 11 reports OLS estimates linking post-reform female inclusion under a manager to the manager's wartime exposure to female coworkers. Columns 1–3 model the probability that a manager includes at least one woman. A 10 percentage point increase in exposure raises this probability by 4.2 points using the maximum exposure measure (column 1) and by 7.2 points using the mean exposure measure (column 2). Column 3 restricts the sample to departmental managers (kacho); the estimate is positive but imprecise. Columns 4–6 report the extensive margin: a 10 percentage point increase in exposure is associated with 0.1–0.2 additional women in the manager's office unit, depending on the measure. The first-stage relationship and IV results are discussed in the next subsection.

5.9 The Causal Effect of Exposure on Post-Reform Female Assignment

Table 12 compares OLS and IV estimates of how wartime exposure to female coworkers affects the post-reform assignment of women to offices. The OLS coefficient is positive and significant, consistent with Table 11. The IV estimate (Panel A, column 2) points in the same direction and is larger in magnitude (5.03 vs. 1.73) but only marginally significant ($p < 0.10$), with a weak first-stage F-statistic (3.19). Columns 3–4 restrict to managers with wartime exposure; the IV estimate remains positive but the first-stage strength collapses (F = 0.07). The results do not allow rejection of a zero causal effect; interpretation must emphasize limited statistical power rather than a precise null.

6 Conclusion

This paper examines the long-term effects of gender diversification in the workplace, leveraging a unique historical shock: the mass conscription of male workers during World War II in Tokyo. Using a novel, detailed personnel dataset, we explore how the forced entry of women into public service roles during wartime influenced gender dynamics and organizational structures in the years that followed.

Our findings reveal three key insights. First, wartime drafting disrupted existing workplace gender compositions, compelling organizations to rely on female employees. This resulted in a measurable and persistent increase in female representation within affected offices, even after the war ended. Second, our instrumental variable estimates demonstrate that exposure to higher female shares during the war significantly influenced long-term gender diversity within organizations. These effects likely stem from changes in attitudes toward gender integration and the normalization of female participation in professional environments.

Third, the benefits of diversification were unevenly distributed. Female employees remained concentrated in lower-tier positions and faced limited opportunities for advancement. Additionally, structural reversals occurred post-war as men reclaimed their positions, highlighting the fragility of gains achieved under temporary shocks.

By demonstrating the causal relationship between exposure to female coworkers and subsequent workplace diversification, this study contributes to the literature on labor markets, gender inequality, and the economics of organizational behavior. Our results emphasize the potential for temporary shocks to drive persistent changes in workplace diversity, while also underscoring the role of institutional and societal constraints in shaping these outcomes.

Table 1: Summary Statistics

Panel A: Sample Counts

	Full Sample	Drafted Years	1944
Unique workers	92,439	54,254	9,323
Unique kyoku	119	48	8
Unique ka (ka × kyoku)	4,967	1,464	276
Unique kakari (kakari × ka × kyoku)	9,234	1,690	383
Unique women	5,968	3,720	702
Workers at rank 3 (shiji/gishi)	15,296	4,989	470
Workers at rank 2 (shoki/gite)	124,127	74,352	8,013
Workers at rank 1 (yatoi/shokutaku)	99,555	29,312	789

Panel B: Kakari-Level Averages

	Full Sample	Drafted Years	1944
Workers per kakari	22.7 (102.4)	58.3 (206.6)	24.3 (38.7)
Unique positions per kakari	2.2 (1.7)	3.4 (2.6)	3.4 (2.3)
Workers per rank-3 role	3.5 (22.5)	5.3 (47.2)	2.3 (3.8)
Workers per rank-2 role	12.8 (48.5)	19.3 (66.1)	8.5 (14.7)
Workers per rank-1 role	11.5 (36.1)	23.6 (67.4)	6.3 (8.3)

Notes: Panel A reports unique counts for the full sample, the subset of years with any drafted workers (1938, 1940, 1941, 1943, 1944), and 1944 alone. Rank 3 = *shiji*/*gishi* (supervisory); Rank 2 = *shoki*/*gite* (clerical/technical staff); Rank 1 = *yatoi*/*shokutaku* (temporary/contract). Panel B reports kakari-level averages (standard deviations in parentheses). A kakari is the lowest organizational unit (team), nested within ka (section) and kyoku (bureau).

Table 2: Balance Test: Pre-War Characteristics (1937–1942) by Future Drafting Status (1944)

Panel A: Unconditional Comparison

	Drafted in 1944		Not Drafted			
	Mean	SD	Mean	SD	Diff	p-value
Female Share	0.027	0.106	0.172	0.305	-0.145	<0.001
New Hire Share	0.335	0.366	0.152	0.258	0.183	<0.001
Promoted Share	0.021	0.105	0.017	0.093	0.004	0.754
Ext. Transfer Share	0.217	0.308	0.228	0.343	-0.011	0.744
N Workers	38.682	120.591	27.200	72.530	11.482	0.163
Avg Rank	2.154	0.833	2.052	0.577	0.102	0.099
Offices	28		21			
Obs (kakari × pos × year)	512		135			

Panel B: Conditional Comparison (Ka + Occupation + Year FE)

	Coef	SE	t-stat	p-value	N
Female Share	0.0018	0.0166	0.11	0.916	626
New Hire Share	—	—	—	—	—
Promoted Share	-0.0251	0.0232	-1.08	0.287	319
Ext. Transfer Share	-0.0843	0.2199	-0.38	0.703	626
N Workers	9.7028	16.7093	0.58	0.564	626
Avg Rank	-0.0837	0.0675	-1.24	0.221	626

Notes: Sample restricted to offices observable in both the pre-war period (1937–1942) and in 1944. Unit of observation: kakari × position × year, matching the regression panel. Panel A reports unconditional means split by whether the office experienced any military drafting in 1944. Panel B reports coefficients from OLS regressions of each characteristic on a drafted-in-1944 indicator, with ka, occupation, and year fixed effects. Standard errors clustered at the office level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 3: Worker Reallocation: Transfers In

Model:	Baseline (1)	$\times$ Rank (2)	$\times$ Eng. (3)	$\times$ Gender (4)
Variables				
No. drafted workers	0.1466**	0.1341***	0.2079	0.1741***
	(0.0599)	(0.0359)	(0.1440)	(0.0619)
No. drafted $\times$ Rank 1		0.1538		
		(0.2452)		
No. drafted $\times$ Rank 3		0.8616**		
		(0.4210)		
No. drafted $\times$ Engineer			-0.0865	
			(0.1356)	
Fixed-effects				
year_num	Yes	Yes	Yes	Yes
ka_id	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
Fit statistics				
Observations	7,409	7,409	7,409	7,409

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: position $\times$ kakari $\times$ year (1938–1945). Dependent variable: number of workers who transferred in from a different bureau group. Transfers are defined as workers whose normalized bureau (kyoku) group changed from the prior year, using a crosswalk that accounts for the 1943 Tokyo-Fu/Shi to Tokyo-To merger. Column (2) interacts the treatment with rank indicators (rank 1 = *yatoi*/*shokutaku*; rank 3 = *shuji*/*gishi*; rank 2 is the omitted category). Column (3) interacts with an engineer indicator. Column (4) interacts with the position's female employment share. Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 4: Transfer Decomposition by Organizational Distance

	Same ka & kyoku			Same kyoku, diff. ka			Different kyoku		
	Base	× Rank	× Eng.	Base	× Rank	× Eng.	Base	× Rank	× Eng.
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
No. drafted workers	-0.0205	-0.0134	-0.0419	0.2910***	0.3462***	0.1492	0.1466**	0.1341***	0.2079
	(0.0148)	(0.0156)	(0.0281)	(0.0838)	(0.0722)	(0.1336)	(0.0599)	(0.0359)	(0.1440)
No. drafted × Rank 1		-0.0722			-0.6017***			0.1538	
		(0.0497)			(0.1517)			(0.2452)	
No. drafted × Rank 3		0.4560*			1.047**			0.8616**	
		(0.2421)			(0.5159)			(0.4210)	
No. drafted × Engineer			0.0302			0.1998			-0.0865
			(0.0345)			(0.1372)			(0.1356)
Fixed-effects									
year_num	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ka_id	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics									
Observations	7,409	7,409	7,409	7,409	7,409	7,409	7,409	7,409	7,409

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: position × kakari × year (1938–1945). Dependent variable: number of workers arriving from each organizational distance. Columns (1)–(3): workers who transferred from a different kakari within the same ka and kyoku (bureau) group. Columns (4)–(6): workers from the same kyoku group but a different ka (section). Columns (7)–(9): workers from a different kyoku group entirely. Kyoku groups use a crosswalk that accounts for the 1943 Tokyo-Fu/Shi to Tokyo-To merger. Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 5: Retention of Non-Drafted Workers

Dep. var: Retained (=1 if worker appears in year t)				
	Baseline	$\times$ Gender	$\times$ Rank	$\times$ Engineer
Model:	(1)	(2)	(3)	(4)
Variables				
No. drafted workers	0.0091	0.0090	0.0129	0.0128
	(0.0062)	(0.0058)	(0.0098)	(0.0085)
No. drafted $\times$ Female		0.0021		
		(0.0139)		
No. drafted $\times$ Rank 1			-0.0089	
			(0.0091)	
No. drafted $\times$ Engineer				-0.0119
				(0.0090)
Female		-0.0080		
		(0.0060)		
Fixed-effects				
year_num	Yes	Yes	Yes	Yes
Ka	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
Fit statistics				
Observations	190,187	190,187	190,187	190,187

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS linear probability model. Dependent variable: retained (=1 if worker appears in year t). Treatment: number of male workers drafted from the same office $\times$ position in year t. All four columns use the full sample of non-drafted workers observed in any year from 1937 to $t-1$, using their most recent observation for office and position assignment. Column 1 estimates the baseline effect. Column 2 interacts the treatment with a female indicator. Column 3 interacts with rank dummies (Rank 1 = low-rank/temporary; Rank 3 = high-rank; Rank 2 is the omitted category). Column 4 interacts with an engineer-position indicator. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 6: Bonus Salary (*Rinji Tokubetsu*) Response to Drafting

Dep. var: No. workers receiving rinji tokubetsu

Model:	Baseline (1)	$\times$ Rank (2)	$\times$ Engineer (3)	$\times$ Gender (4)
Variables				
No. drafted $\times$ Female share				-0.4492
				(0.3928)
No. drafted workers	0.1361**	0.1538***	0.0230	0.1569**
	(0.0522)	(0.0578)	(0.0430)	(0.0696)
No. drafted $\times$ Rank 1		-0.1806***		
		(0.0614)		
No. drafted $\times$ Rank 3		0.2938		
		(0.1793)		
No. drafted $\times$ Engineer			0.1481**	
			(0.0744)	
Female share				1.854*
				(0.9597)
Fixed-effects				
ka_id	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
Fit statistics				
Observations	576	576	576	576

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regression using 1944 cross-section. Unit: position $\times$ office. Dependent variable: number of non-drafted workers receiving *rinji tokubetsu* (temporary special allowance). Treatment: number of male workers drafted from the same office $\times$ position. Column (1) is the baseline specification. Column (2) interacts the treatment with rank indicators (Rank 2 = regular civil servant is the base; Rank 1 = *yatoi*/*shokutaku*; Rank 3 = *shuji*/*gishi*). Column (3) interacts with an engineer indicator. Column (4) interacts with the female share of workers in the position. All specifications include ka and position fixed effects and control for log cumulative male baseline. Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 7: Short-Run Promotion in Response to Drafting

Dep. var: Promoted (=1 if rank increased from $t-1$ to t)	Baseline	$\times$ Gender	$\times$ Rank	$\times$ Engineer
Model:	(1)	(2)	(3)	(4)
Variables				
No. drafted workers (office)	0.0881**	0.0891**	0.0820**	0.0923**
	(0.0400)	(0.0399)	(0.0356)	(0.0381)
No. drafted $\times$ Female		0.0279***		
		(0.0079)		
Female		-0.0220		
		(0.0183)		
No. drafted $\times$ Rank 1			0.0353	
			(0.0516)	
No. drafted $\times$ Rank 3			-0.0240	
			(0.0268)	
No. drafted $\times$ Engineer				-0.0425***
				(0.0111)
Fixed-effects				
year_num	Yes	Yes	Yes	Yes
Ka	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
Fit statistics				
Observations	8,754	8,754	8,754	8,754

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions (linear probability model). Dependent variable: promoted (=1 if worker's position rank increased from year $t-1$ to t). Sample: non-drafted workers present in the same office in consecutive years. Treatment: total number of male workers drafted from the same office in year t (office-level variation). Column 2 interacts treatment with a female indicator; Column 3 interacts with rank indicators (rank 2 is the omitted category; rank 1 = temporary *yatoi/shokutaku*, rank 3 = supervisory *shiji/gishi*); Column 4 interacts with an engineer indicator (positions containing UTF8min). Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 8: New Hiring in Response to Military Drafting

	No. new hires			No. new female			No. new male		
Model:	Baseline (1)	× Rank (2)	× Eng. (3)	All (4)	Rank 1 (5)	Rank 2+ (6)	All (7)	Rank 1 (8)	Rank 2+ (9)
Variables									
No. drafted workers	0.2171**	0.2513**	0.0959	0.0073	-0.0047	0.0120**	0.2126	-0.0680	0.2806**
	(0.1091)	(0.1042)	(0.0869)	(0.0061)	(0.0042)	(0.0055)	(0.1372)	(0.0420)	(0.1104)
No. drafted × Rank 1		-0.3642**							
		(0.1444)							
No. drafted × Rank 3		1.129**							
		(0.5364)							
No. drafted × Engineer			0.1709						
			(0.1486)						
Fixed-effects									
year_num	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ka_id	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics									
Observations	7,409	7,409	7,409	7,409	7,409	7,409	7,409	7,409	7,409

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: position × kakari × year (1938–1945). Columns (1)–(3): dependent variable is total new hires. Columns (4)–(6): new female hires (all, rank 1, rank 2+). Columns (7)–(9): new male hires (all, rank 1, rank 2+). Column (2) interacts the treatment with rank indicators (rank 1 = *yatoi*/*shokutaku*; rank 3 = *shuji*/*gishi*; rank 2 is the omitted category). Column (3) interacts with an engineer indicator. Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 9: Gender Composition and Tenure

Dep. var:	Female share			Avg. tenure		
	Baseline	× Rank	× Eng.	Baseline	× Rank	× Eng.
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
No. drafted workers	0.0006	0.0005	0.0026	-0.0373***	-0.0275**	-0.0793***
	(0.0009)	(0.0009)	(0.0021)	(0.0116)	(0.0114)	(0.0216)
No. drafted × Rank 1		0.0015			-0.0800*	
		(0.0026)			(0.0470)	
No. drafted × Rank 3		-0.0122			-0.2095	
		(0.0132)			(0.1490)	
No. drafted × Engineer			-0.0029			0.0614**
			(0.0022)			(0.0244)
Fixed-effects						
year_num	Yes	Yes	Yes	Yes	Yes	Yes
Ka	Yes	Yes	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	6,593	6,593	6,593	6,593	6,593	6,593

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions at the position × year level (1938–1945). Columns (1)–(3): dependent variable is female share of all workers. Columns (4)–(6): dependent variable is average tenure (years in data). Column (1) and (4) report baseline effects. Columns (2) and (5) interact draft exposure with rank indicators (rank 1 = *yatoi*/temporary; rank 3+ = senior; rank 2 = base category, absorbed by position FE). Columns (3) and (6) interact draft exposure with an engineer indicator (positions containing UTF8min). All specifications include log(cumulative male baseline + 1) as a control (not displayed), as well as year, ka, and position fixed effects. Standard errors clustered at the office level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 10: Postwar Outcomes of 1944 Colleagues (1946–1955)

Dependent Variables:	rank_z_change Promotion		avg_female_share_dest Dest. F.Share	
Model:	(1)	(2)	(3)	(4)
Variables				
Draft share	-0.0919	-0.0929	-0.0018	-0.0018
	(0.1438)	(0.1457)	(0.0070)	(0.0059)
Female (=1)		-0.2087		0.1283***
		(0.1482)		(0.0477)
Draft share × Female		0.0047		0.0310
		(0.7866)		(0.1611)
Fixed-effects				
ka_id_1944	Yes	Yes	Yes	Yes
pos_norm_1944	Yes	Yes	Yes	Yes
Fit statistics				
Observations	1,281	1,281	1,281	1,281
R^2	0.46085	0.46163	0.09966	0.24099

Clustered (office_id_1944) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Sample: non-drafted workers observed in 1944. Columns (1)–(2): dependent variable is z-scored rank change (postwar avg. minus 1944 rank, standardized within era). Column (2) includes gender interaction. Columns (3)–(4): dependent variable is average female share of the worker's office-position across postwar years. Column (4) includes gender interaction. All specifications include ka and position fixed effects from 1944 and control for log cumulative male stock. Standard errors clustered at the office level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 11: Effect of Military Drafting on Remaining Office Size

Model:	Baseline (1)	× Rank (2)	× Eng. (3)	× Gender (4)
Variables				
No. drafted workers	0.9225***	1.120***	1.385*	0.7134***
	(0.2762)	(0.3167)	(0.7122)	(0.2369)
No. drafted × Rank 1		-2.159***		
		(0.6828)		
No. drafted × Rank 3		8.553***		
		(3.112)		
No. drafted × Engineer			-0.6566	
			(0.6895)	
Fixed-effects				
year_num	Yes	Yes	Yes	Yes
ka_id	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
Fit statistics				
Observations	7,409	7,409	7,409	7,409

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: position × kakari × year (1938–1945). Dependent variable: remaining headcount (total workers minus drafted workers). Column (2) interacts the treatment with rank indicators (rank 1 = *yatoi*/*shokutaku*; rank 3 = *shuji*/*gishi*; rank 2 is the omitted category). Column (3) interacts with an engineer indicator. Column (4) interacts with the position's female employment share. Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 12: Characteristics of Donor Offices (Offices with Transfers Out)

Panel A: Unconditional Comparison

	Donor		Non-Donor			
	Mean	SD	Mean	SD	Diff	p-value
Has New Hire (=1)	0.769	0.427	0.613	0.487	0.156	0.028
No. New Hires	15.949	22.369	5.262	14.413	10.687	0.005
Avg Tenure (years)	2.452	0.901	2.559	1.559	-0.106	0.469
Female Share	0.037	0.095	0.044	0.142	-0.007	0.654

Panel B: Conditional Comparison (Ka + Position + Year FE)

	Coef	SE	t-stat	p-value	N
Has New Hire (=1)	0.1197*	0.0695	1.72	0.085	7,409
No. New Hires	8.1539*	4.8527	1.68	0.093	7,409
Avg Tenure (years)	-0.0715	0.1131	-0.63	0.527	7,409
Female Share	0.0037	0.0212	0.17	0.862	7,409

Notes: Sample: kakari $\times$ position $\times$ year (1938–1945). A *donor* office is one that had at least one worker transfer out in that year. Panel A reports unconditional means. Panel B reports coefficients from OLS regressions of each characteristic on a donor indicator, with ka, position, and year fixed effects. Standard errors clustered at the office level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 13: Characteristics of Transferred Workers

Model:	Female (1)	Tenure (2)	Rank (3)	All (4)
Variables				
Female (=1)	0.0196			0.0200
	(0.0142)			(0.0136)
Tenure (years)		0.0008		0.0012
		(0.0011)		(0.0011)
Rank (enriched)			-0.0104	-0.0112
			(0.0114)	(0.0114)
Fixed-effects				
year_num	Yes	Yes	Yes	Yes
lag_ka_id	Yes	Yes	Yes	Yes
lag_pos	Yes	Yes	Yes	Yes
Fit statistics				
Observations	49,036	49,036	49,036	49,036

Clustered (lag_office) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions (linear probability model). Dependent variable: transferred (=1 if worker moved to a different unit from prior year). Unit of observation: individual worker × year (1938–1945), restricted to workers with prior-year records. Rank is an enriched 1–5 scale serving as a productivity proxy. All specifications include year, ka (prior-year), and position (prior-year) fixed effects. Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

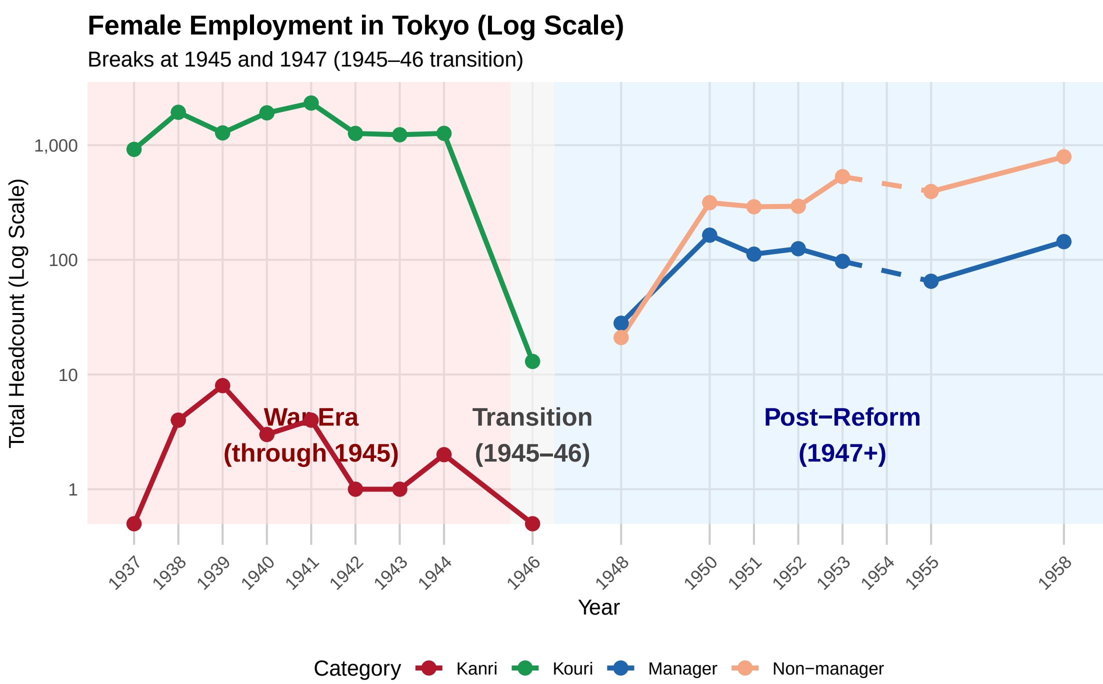
Table 14: Postwar Outcomes of Remaining Workers in Donor vs. Non-Donor Offices

	Tenure		Promotion	
Model:	(1)	(2)	(3)	(4)
Variables				
Donor office (=1)	-0.2545	-0.2152	-0.3553***	-0.3612***
	(0.2147)	(0.2243)	(0.0860)	(0.0856)
Female (=1)		-0.5768***		-0.2237*
		(0.0932)		(0.1216)
Donor × Female		-0.3234*		
		(0.1887)		
Fixed-effects				
ka_id	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
Fit statistics				
Observations	3,385	3,385	1,276	1,276
R^2	0.09040	0.09689	0.42694	0.42793

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

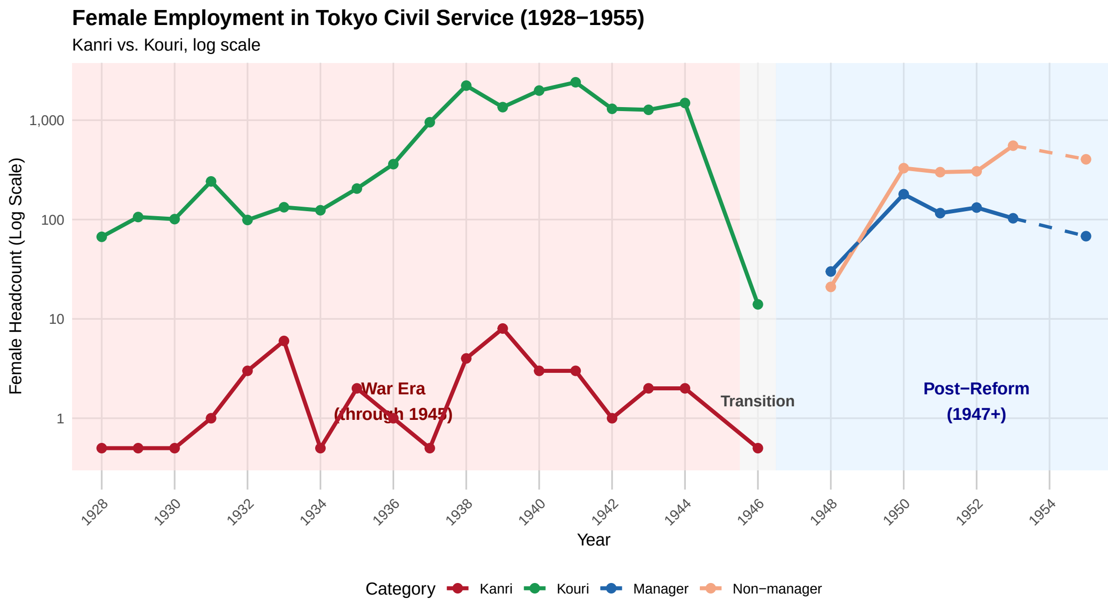
Notes: OLS regressions. Sample: non-drafted workers observed in 1944. Columns (1)–(2): dependent variable is postwar tenure (count of years appearing 1946–1955). Columns (3)–(4): dependent variable is z-scored rank change (postwar avg. minus 1944 rank). A *donor* office had at least one worker transfer out during 1938–1945. The comparison group is same-ka offices (kakari × position) that did not have transfers out. All specifications include ka and position fixed effects from 1944 and control for log cumulative male stock (not shown). Standard errors clustered at the office level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure 1: Female Employment in Tokyo by Rank Category



Note: Headcounts are shown on a log scale; the gap reflects the 1946-1949 transition period.

Figure 2: Women in the Civil Service, 1928–1955



Note: Counts are shown on a log scale. Solid lines indicate observed data; dashed segments connect across gaps in the source records. *Kanri* () denotes appointed officials; *Kouri* () denotes public employees.

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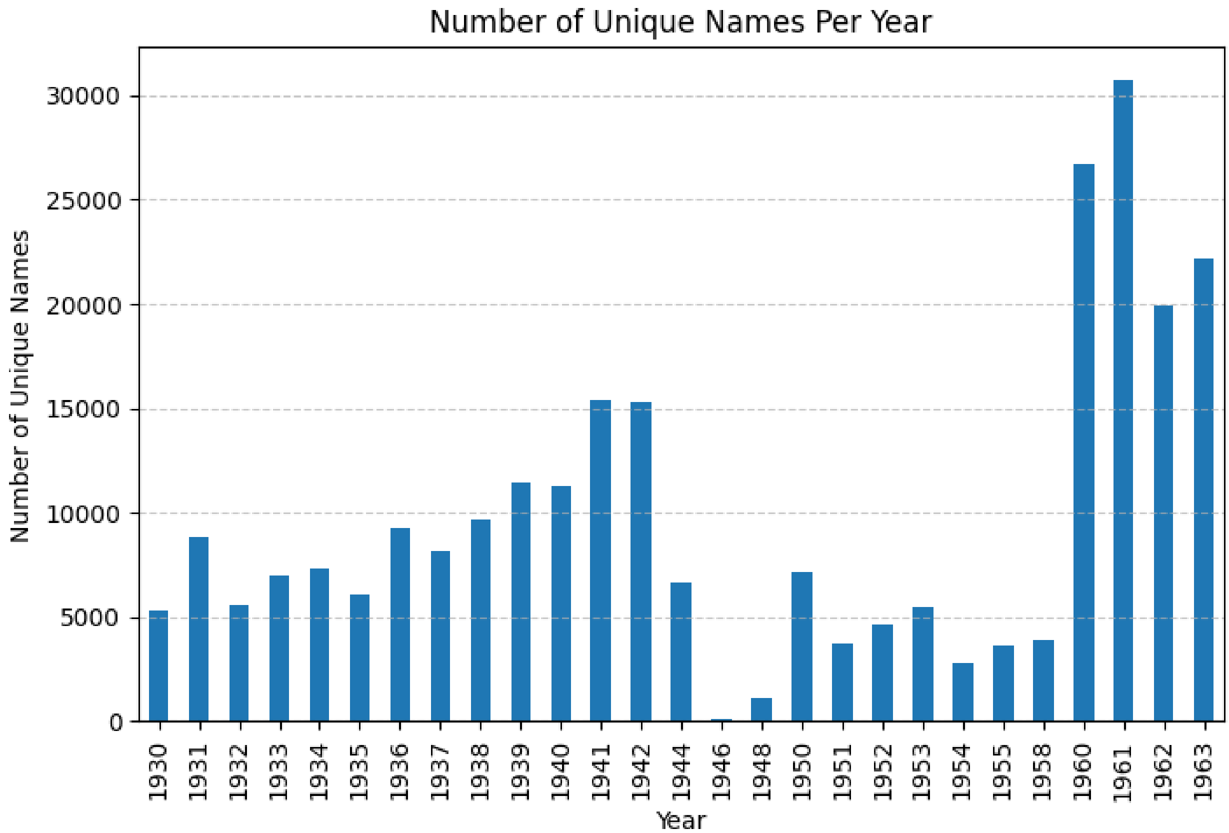
Figure 3: Number of unique names by year



A Appendix

Table 15: Wartime Female Exposure and Postwar Female Inclusion in Kakari

	Has Female (Binary)			Number of Females		
	(1) Kakaricho (Max)	(2) Kakaricho (Mean)	(3) Kacho (Max)	(4) Kakaricho (Max)	(5) Kakaricho (Mean)	(6) Kacho (Max)
Kakaricho Exposure (Max)	0.418***			1.165**		
	(0.092)			(0.374)		
Kakaricho Exposure (Mean)		0.715***			1.810**	
		(0.141)			(0.563)	
Kacho Exposure (Max)			0.242			−0.485
			(0.200)			(0.743)
Engineer Share (Ka)	0.015	0.019	0.309***	−0.381**	−0.374**	−0.449
	(0.040)	(0.040)	(0.091)	(0.136)	(0.137)	(0.637)
No. Kakaricho (Ka)	0.018***	0.018***	0.016***	−0.036	−0.036	−0.096*
	(0.003)	(0.003)	(0.004)	(0.027)	(0.027)	(0.045)
Kakari Size	0.006***	0.006***	0.005***	0.061***	0.061***	0.076***
	(0.001)	(0.001)	(0.001)	(0.014)	(0.014)	(0.014)
Kyoku FE	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Office	Office	Office	Office	Office	Office
Observations	2,324	2,324	613	2,324	2,324	613
R^2 (within)	0.174	0.177	0.247	0.502	0.502	0.562

Notes: Columns 1–3 estimate a linear probability model $\mathbf{1}[\text{Female}_{j,t} > 0] = \beta\,\text{Exposure}_{m(j)} + X'_{j,t}\gamma + \iota_k + \varepsilon_{j,t}$; columns 4–6 estimate $N^f_{j,t} = \beta\,\text{Exposure}_{m(j)} + X'_{j,t}\gamma + \iota_k + \varepsilon_{j,t}$ by OLS. The unit of observation is kakari $\times$ year (postwar, 1947–). Exposure is the share of females in the manager $m(j)$'s wartime (1937–1945) position $\times$ kakari $\times$ year cell, aggregated as maximum (cols. 1, 3, 4, 6) or mean (cols. 2, 5) across wartime years. Kakaricho = subsection chief; Kacho = section chief. Controls include engineer share, number of kakaricho, and kakari size. Standard errors clustered at office level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Second Stage (Dep. var.: Number of Females)

Table 16: Direct Draft Shock: OLS vs. IV

	OLS (Uncond) (1)	IV (Uncond) (2)	OLS (Cond) (3)	IV (Cond) (4)
Exposure (uncond)	1.730***	5.033*		
	(0.133)	(2.771)		
Exposure (cond)			1.266***	3.846
			(0.241)	(16.56)
Engineer share	0.047**	0.029	−0.043	0.025
	(0.019)	(0.026)	(0.075)	(0.447)
No. Kakari-cho	−0.001	−0.005	0.006	0.003
	(0.009)	(0.009)	(0.009)	(0.022)
Kakari size	0.030***	0.031***	0.027***	0.028***
	(0.005)	(0.004)	(0.006)	(0.008)
Year FE	Yes	Yes	Yes	Yes
Kyoku FE	Yes	Yes	Yes	Yes
Observations	6,662	6,662	2,324	2,324
R^2	0.275	0.207	0.341	0.298
F-test (1st stage)		3.187		0.074

Panel B: First Stage (Dep. var.: Exposure)

	OLS (Uncond) (1)	IV (Uncond) (2)	OLS (Cond) (3)	IV (Cond) (4)
Draft shock (uncond)		−0.029***		
		(0.010)		
Draft shock (cond)				−0.005
				(0.019)
Engineer share		0.003		−0.021
		(0.008)		(0.013)
No. Kakari-cho		0.001***		0.001*
		(0.000)		(0.001)
Kakari size		−0.000***		−0.000
		(0.000)		(0.000)
Year FE		Yes		Yes
Kyoku FE		Yes		Yes
Position FE		Yes		Yes
Observations		6,661		2,324
R^2		0.079		0.064

Notes: Panel A reports estimates of $Y_{j,t} = \beta\,\text{Exposure}_j + X'_{j,t}\gamma + \iota_k + \lambda_t + \varepsilon_{j,t}$, where Y is the number of females in kakari j in postwar year t (1947–) and Exposure is the mean share of females in the position $\times$ kakari $\times$ year cell during wartime (1937–1945). Columns 1–2 are unconditional OLS and IV; columns 3–4 are conditional OLS and IV. IV specifications instrument Exposure with the wartime draft shock (share of males drafted); Panel B reports the corresponding first stage in columns 2 and 4. "Cond" conditions on kakaricho (subsection chief) wartime offices; "Uncond" uses all wartime survivors. Controls include engineer share, number of kakaricho, and kakari size. Standard errors clustered at the office level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 17: New Hiring in Response to Adjacent Military Drafting

Model:	Own Drafts (1)	Adj: Diff Kakari (2)	Adj: Diff Occ (3)
Variables			
No. drafted workers	0.6736***		
	(0.1172)		
Adj. drafts (diff kakari, same occ)		0.0746	
		(0.0718)	
Adj. drafts (diff occ, same kakari)			0.0062
			(0.0942)
Fixed-effects			
year_num	Yes	Yes	Yes
Ka	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes
Fit statistics			
Observations	7,409	7,409	7,409

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Unit of observation: position $\times$ kakari $\times$ year (1938–1945). OLS regressions. Dependent variable: number of new hires. Column (1) uses own-position drafts as treatment. Column (2) uses drafts at different kakari within the same occupation. Column (3) uses drafts at different occupation within the same kakari. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline. Standard errors clustered at the office level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 18: Worker Reallocation in Response to Military Drafting (1944 Only)

Dependent Variables:	n_transfers_in	n_transfers_same_ka	n_transfers_diff_ka	n_transfers_dist2	n_tran
	Transfers	Same Ka	Diff Ka	Same Kyoku	Dif
	In	Transfers	Transfers	Transfers	Tr
Model:	(1)	(2)	(3)	(4)	
Variables					
Own drafts (male)	0.0367***	0.0024	0.0368***	0.0274**	0.0
	(0.0038)	(0.0119)	(0.0038)	(0.0112)	(0
log(Male baseline + 1)	-0.0927		-0.0920	-8.800***	-0
	(0.1038)		(0.1035)	(0.4304)	(0
Adj. drafts (diff kakari, same occ)					
Adj. drafts (diff occ, same kakari)					
Fixed-effects					
kyoku	Yes	Yes	Yes	Yes	
pos_norm	Yes	Yes	Yes	Yes	
Fit statistics					
Observations	802	60	802	184	

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Poisson regressions restricted to 1944. Unit of observation: position × kakari. Column specifications mirror Table 3. All specifi errors clustered at the office level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 19: Gender Composition of Hiring and Workforce (1944 Only)

Dependent Variables:	n_new_female	n_new_male	n_new_female	n_new_male	female_share_all		
	Female Hires	Male Hires	Female Hires	Male Hires	Fem. Share (All, R1)	Fem. Share (All, R2)	F (
Model:	(1)	(2)	(3)	(4)	(5)	(6)	
	Poisson	Poisson	Poisson	Poisson	OLS	OLS	
Variables							
Own drafts (male)	0.0382**	0.0353***	0.0374**	0.0338***			
	(0.0179)	(0.0066)	(0.0187)	(0.0067)			
log(Male baseline + 1)	-0.4911***	0.2378***	-0.5366***	0.2069**			
	(0.1375)	(0.0731)	(0.1876)	(0.0836)			
Own drafts $\times$ Rank 1			0.0181	0.0530***			
			(0.0499)	(0.0198)			
Any drafted (=1)					-0.3113	-0.0115	
					(0.2051)	(0.0142)	
Male baseline					-7.31×10^{-5}	-0.0009*	$-$1
					(7.13×10^{-5})	(0.0005)	(2.
Fixed-effects							
ka	Yes	Yes	Yes	Yes	Yes	Yes	
pos_norm	Yes	Yes	Yes	Yes	Yes	Yes	
Fit statistics							
Observations	232	370	232	370	45	1,246	

Clustered (office_id) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Restricted to 1944. Column specifications mirror Table 9. Poisson regressions for hire counts; OLS for female shares. All position fixed effects. Standard errors clustered at the office level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 20: Postwar Retention of 1944 Colleagues (1946–1955)

Dependent Variable:	tenure_postwar	
	Retention	
Model:	(1)	(2)
Variables		
Draft share	-0.1288	-0.0419
	(0.2345)	(0.2399)
Female (=1)		-0.5192***
		(0.1307)
Draft share × Female		-0.3388
		(0.4196)
Fixed-effects		
ka_id_1944	Yes	Yes
pos_norm_1944	Yes	Yes
Fit statistics		
Observations	3,398	3,398
R^2	0.08863	0.09526

Clustered (office_id_1944) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Sample: non-drafted workers observed in 1944. Dependent variable is count of postwar years (1946–1955) appearing in civil service. Column (2) includes gender interaction. All specifications include ka and position fixed effects from 1944 and control for log cumulative male stock. Standard errors clustered at the office level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.